

**Chennai Enquiry Bot using Retrieval-Augmented
Generation (RAG) and Local LLM (Ollama)**

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1. Introduction

In the age of artificial intelligence, chatbots are becoming an essential tool for providing instant information and assistance to users. They simulate human conversation and offer fast, accurate responses to queries in natural language. From customer service to travel guides, chatbots are transforming the way people access information.

This project focuses on developing a **Chennai Enquiry Bot**, a domain-specific chatbot designed to serve both **tourists visiting Chennai** and **locals living in the city**. Chennai is a metropolitan hub, famous for its beaches, temples, food, culture, and shopping, making it an ideal candidate for such a system. The bot can answer questions about **tourist attractions, food, shopping, movies, transport, weather, and culture**.

Unlike traditional chatbots that rely only on predefined rules and keyword matching, this system is enhanced using **Retrieval-Augmented Generation (RAG)** combined with a **local Large Language Model (LLM)** through **Ollama**. This integration makes the chatbot capable of giving **context-aware, natural, and human-like answers** while running completely offline, without relying on paid APIs or cloud services.

2. Objectives

The main objectives of the project are:

1. To build a conversational AI system that helps both **tourists and locals** in Chennai.
2. To define **intents** (user goals) and **entities** (important keywords) that cover the most common queries.
3. To use **Python** for implementation, combining **embeddings for retrieval** and **LLM for answer generation**.
4. To design the system with an **external JSON knowledge base** so that information can be easily updated and expanded.
5. To add a **memory feature** that allows the chatbot to remember the user's name, making the conversation more personal.
6. To run the chatbot using **Ollama** for offline and cost-free deployment.

3. System Architecture

The Chennai Enquiry Bot is built using three major layers:

1. **Knowledge Base Layer**
 - Stores structured information in a JSON file.
 - Contains details on Chennai's places, food, shopping, movies, transport, weather, and culture.

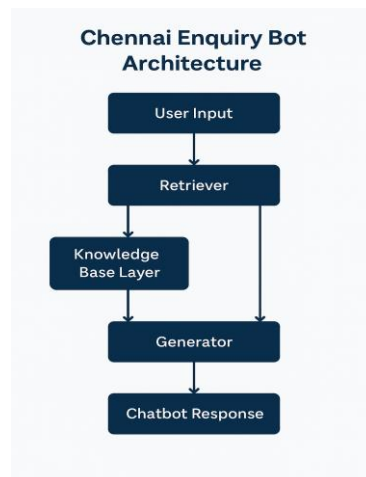
2. Retriever Layer

- Uses **Sentence Transformer (all-MiniLM-L6-v2)** to convert both queries and knowledge base entries into embeddings (vector representations).
- Employs **cosine similarity** to match user queries with the closest entries in the knowledge base.

3. Generator Layer (LLM via Ollama)

- The retrieved information along with the user's query is passed to an **LLM (Mistral model via Ollama)**.
- The model generates fluent, polite, and contextually relevant responses.

Architecture Flow



4. Workflow of the Chatbot

1. **User Query:** The user asks a question, such as *“What food is Chennai famous for?”*.
2. **Retrieval:** The chatbot converts the query into an embedding and compares it with embeddings of the JSON knowledge base.
3. **Context Selection:** The top matching entries are chosen as the most relevant context.
4. **LLM Processing:** The context and query are given to the LLM, which produces a well-formed answer.
5. **Personalization:** If the user introduces their name, the chatbot stores it and recalls it later.

6. **Fallback Handling:** If the system cannot confidently match a query, it politely asks the user to reframe or suggests available categories.

5. Features of the Chennai Enquiry Bot

1. **Domain-Specific Guidance:** Answers questions about Chennai across multiple categories.
2. **Retrieval-Augmented Generation:** Combines structured knowledge with generative AI for flexible responses.
3. **Natural Language Output:** Provides smooth, conversational answers instead of robotic or predefined text.
4. **JSON Knowledge Base:** Keeps content separate from code, making updates simple.
5. **Personalization:** Learns and remembers the user's name during the conversation.
6. **Fallback Responses:** Politely handles queries outside its knowledge base.
7. **Offline and Free:** Runs entirely on local hardware using Ollama, requiring no internet connectivity or API payments.

6. Challenges Faced and Solutions

Challenge	Description	Solution Implemented
API cost and quota issues	Using OpenAI GPT required paid API and credits	Shifted to Ollama to run free, offline models locally
Handling diverse queries	Users can ask the same thing in many ways	Used embeddings and similarity search to capture meaning instead of relying on exact words
Updating responses	Hardcoding answers in Python would make updates difficult	Stored information in external JSON file for easy maintenance
Personalization	Users expect the chatbot to feel interactive and human-like	Added memory component to store and recall user's name
Speed and performance	Large models like LLaMA 2 (13B) are resource-heavy	Chose Mistral model (smaller and faster) to balance accuracy and performance

7. Implementation

chennai_knowledge.json

```
{  
    "places": "Top tourist places in Chennai include Marina Beach, Kapaleeshwarar Temple, Santhome Cathedral, Fort St. George, and Mahabalipuram nearby.",  
    "food": "Famous foods in Chennai are dosa, idli, filter coffee, Chettinad chicken, and Chennai-style biryani.",  
    "shopping": "For shopping, visit T. Nagar for sarees and jewelry, Pondy Bazaar for street shopping, and Express Avenue or Phoenix Mall for brands.",  
    "movies": "Sathyam Cinemas and PVR in Phoenix Mall are the best theatres to watch Tamil and international movies.",  
    "transport": "You can travel around Chennai by metro, MTC buses, suburban trains, and autos.",  
    "weather": "Chennai is hot most of the year. The best time to visit is November to February when the weather is cooler.",  
    "culture": "Chennai is known for Bharatanatyam dance, Carnatic music, and cultural festivals like Pongal and the December Music Season."  
}
```

chennai_bot.py

```
import json  
import numpy as np  
from sentence_transformers import SentenceTransformer  
from sklearn.metrics.pairwise import cosine_similarity  
import ollama  
  
with open("chennai_knowledge.json", "r") as file:  
    documents = json.load(file)  
  
doc_texts = list(documents.values())  
doc_keys = list(documents.keys())
```

```
embed_model = SentenceTransformer("all-MiniLM-L6-v2")
```

```
doc_embeddings = embed_model.encode(doc_texts)
```

```
def retrieve_context(query, top_k=2):
```

```
    query_embedding = embed_model.encode([query])
```

```
    similarities = cosine_similarity(query_embedding, doc_embeddings)[0]
```

```
    top_indices = np.argsort(similarities)[::-1][:top_k]
```

```
    return [doc_texts[i] for i in top_indices]
```

```
def rag_llm_chatbot(query, user_name=None):
```

```
    context = retrieve_context(query)
```

```
    prompt = f"""
```

```
    You are a Chennai Enquiry Bot for tourists and locals.
```

```
    Use the following context to answer naturally:
```

```
    Context: {" ".join(context)}
```

```
    User Query: {query}
```

```
    """
```

```
    response = ollama.chat(
```

```
        model="mistral",
```

```
        messages=[
```

```
            {"role": "system", "content": "You are a helpful Chennai tourist assistant."},
```

```
            {"role": "user", "content": prompt}
```

```
        ]
```

```
    )
```

```
    answer = response['message']['content']
```

```
    if user_name and "hi" in query.lower():
```

```
        answer += f" Nice to see you again, {user_name}!"
```

```
return answer
```

```
def chat():
```

```
    print("CHENNAI Enquiry Bot: Hello! Ask me about places, food, shopping, movies,  
or culture.")
```

```
    user_name = None
```

```
    while True:
```

```
        query = input("You: ")
```

```
        if "bye" in query.lower():
```

```
            print("Bot: Goodbye! Have a great time in Chennai ")
```

```
            break
```

```
        elif "my name is" in query.lower():
```

```
            user_name = query.split("is")[-1].strip().capitalize()
```

```
            print(f"Bot: Nice to meet you, {user_name}! I'll remember your name.")
```

```
            continue
```

```
        elif "what's my name" in query.lower():
```

```
            if user_name:
```

```
                print(f"Bot: Your name is {user_name}.")
```

```
            else:
```

```
                print("Bot: I don't know your name yet. Tell me using 'My name is ...'")
```

```
            continue
```

```
        response = rag_llm_chatbot(query, user_name)
```

```
        print("Bot:", response)
```

```
if __name__ == "__main__":
```

```
    chat()
```

Output and Implementation

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Arjun\AI Chatbot>ollama --version
ollama version is 0.12.6

C:\Arjun\AI Chatbot>ollama pull mistral
pulling manifest
pulling f5074b1221da: 100% 4.4 GB
pulling 43078e2d4e53: 100% 11 MB
pulling 1ff5b64b61b9: 100% 799 B
pulling ed1leda7790d: 100% 30 B
pulling 1964e17101bd: 100% 487 B
verifying sha256 digest
writing manifest
success

C:\Arjun\AI Chatbot>
```

```
C:\Windows\System32\cmd.exe
C:\Arjun\AI Chatbot>python chennai_bot.py
CHENNAI Enquiry Bot: Hello! Ask me about places, food, shopping, movies, or culture.
You: Hi
Bot: Hello! How can I assist you today? I'm here to help with any questions or information about tourist spots, shopping areas, food recommendations, and local experiences in Chennai. Let me know if you need any suggestions for your visit!
You: I want to know about shopping places.
Bot: Hello! If you're interested in shopping during your stay in Chennai, here are some recommendations for you:

1. T. Nagar (Thyagaraya Nagar) is famous for its sarees and jewelry shops. The most popular ones include RmkV Silks, Ven u Silver Collections, and Mandira Sarees.

2. Pondy Bazaar, located in George Town, is a great place for street shopping. You can find a variety of items like text files, electronics, jewelry, and more here at affordable prices.

3. If you're looking for international brands, head to either Express Avenue or Phoenix Mall, both are modern shopping malls with a wide range of shops, restaurants, and entertainment options.

4. For local handicrafts and souvenirs, visit Khadi Gramodyog Emporium in Nungambakkam, which offers traditional Indian textiles and products.
You: what about food
Bot: In Chennai, you'll find a rich culinary scene that offers a variety of delicious dishes. Some of the famous foods you should definitely try include dosa, idli (both are types of rice cakes), filter coffee (traditional South Indian coffee), Chettinad chicken (spicy chicken dish from the Chettinad region), and Chennai-style biryani (a flavorful rice dish made with aromatic basmati rice, meat, and spices).

There are numerous restaurants and street food stalls throughout the city where you can savor these delicacies. Some popular areas for food include T. Nagar, Mylapore, and George Town. Don't forget to ask for recommendations from locals when you visit! Enjoy your culinary journey in Chennai!
```

```
C:\Windows\System32\cmd.exe
You: what about food
Bot: In Chennai, you'll find a rich culinary scene that offers a variety of delicious dishes. Some of the famous foods you should definitely try include dosa, idli (both are types of rice cakes), filter coffee (traditional South Indian coffee), Chettinad chicken (spicy chicken dish from the Chettinad region), and Chennai-style biryani (a flavorful rice dish made with aromatic basmati rice, meat, and spices).

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You: give me about beaches in chennai
Bot: One of the most famous and popular beaches in Chennai is Marina Beach, which stretches over 13 km along the Bay of Bengal. It's one of the longest urban beaches in the world and offers beautiful views, especially during sunrise and sunset. Visitors can enjoy various activities like swimming, horseback rides, and boat rides.

Another notable beach near Chennai is Mahabalipuram Beach, located approximately 60 km south of the city. It's famous for its stunning rock-cut temples that date back to the 7th century, making it a UNESCO World Heritage Site. Besides the historical sites, visitors can also relax on the beach or take part in water activities like swimming and surfing.

Additionally, Besant Nagar Beach, often referred to as Elliot's Beach, is another popular option in Chennai. It offers a more peaceful atmosphere compared to Marina Beach, with cleaner sands and less crowded areas. The beach also has a charming pier and a nearby temple dedicated to Kali Amman, making it an ideal location for exploring both natural beauty and cultural landmarks.

You can easily reach these beaches in Chennai using public transportation like metro, MTC buses, suburban trains, or by hiring auto-rickshaws. For a memorable beach experience during your visit to Chennai, don't miss out on exploring these beautiful coastal locations!
You: bye
Bot: Goodbye! Have a great time in Chennai

C:\Arjun\AI Chatbot>
```

8. Testing

Case 1: Greeting

You: Hi

Bot: Hello! I'm your Chennai Enquiry Bot. Ask me about places, food, shopping, movies, or culture.

Case 2: Food Enquiry

You: What food is Chennai famous for?

Bot: Chennai is famous for dosa, idli, authentic filter coffee, Chettinad chicken, and its unique style of biryani.

Case 3: Shopping Guidance

You: Where can I shop?

Bot: For traditional shopping, T. Nagar and Pondy Bazaar are excellent choices. For malls, you can visit Express Avenue or Phoenix MarketCity.

Case 4: Personalization with Memory

You: My name is Arjun

Bot: Nice to meet you, Arjun! I'll remember your name.

You: What's my name?

Bot: Your name is Arjun.

Case 5: Out-of-Scope Query

You: Do you know about cricket?

Bot: Sorry, I didn't get that. Do you want info on places, food, shopping, or movies?

Case 6: Exit Command

You: Bye

Bot: Goodbye! Have a great time in Chennai

9. Results

- The chatbot provided **accurate and relevant responses** for queries across all categories.
- The LLM generated **human-like and polite answers**, improving user experience.
- The **name memory feature** worked successfully, adding personalization.
- Fallback handling prevented dead-end conversations.
- The system operated **offline and free of cost** using Ollama, demonstrating a practical alternative to API-based bots.

10. Conclusion

The Chennai Enquiry Bot is a successful implementation of a **domain-specific chatbot** powered by **Retrieval-Augmented Generation** and a **local LLM**. It

overcomes the limitations of rule-based systems by providing flexible, contextual, and natural answers. The use of a **JSON knowledge base** makes it scalable and easy to maintain, while the **memory feature** adds a human touch.

Running on **Ollama** ensures that the system is cost-effective and independent of external APIs. This makes it highly practical for real-world applications, especially where internet access or budget is limited.

The project demonstrates how AI can be applied in a **city-specific guide system**, useful for both locals and tourists.